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# ECE 310 Fall 2026
## Lecture 11: BIBO stability and causality

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# Learning Objectives

- Explain the conditions for an LTI system to be BIBO stable using: the definition of BIBO stability, impulse response, or $z$-transform.
- Define what makes a causal and non-causal LTI systems in the $z$-domain.
- Identify bounded inputs that cause unstable systems to yield unbounded outputs.



# BIBO Stability

Thus far, we have given two conditions for BIBO stability of a discrete-time system:

1. Following the definition of BIBO stability: A discrete-time system $T$ is *bounded-input bounded-output stable* (BIBO stable) if and only if for any $x[n]$ such that $|x[n]| < \beta$ for all $n$, $|T(x[n])| < \alpha$ for all $n$ where $0 \leq \alpha < \infty,\ 0 \leq \beta < \infty$.

2. For an LTI system with impulse response $h[n]$: An LTI system is BIBO stable if and only if

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty.$$



# BIBO Stability in the $z$-domain

We can also demonstrate another necessary and sufficient condition for BIBO stability in the $z$-domain. Let $H(z)$ be the transfer function of an LTI system with an ROC that contains the unit circle, i.e. $|z| = 1$. This implies:

$$\begin{aligned}
\infty &> \left( \sum_{n=-\infty}^{\infty} |h[n]z^{-n}| \right) \Big|_{z=e^{j\theta}} \\
&= \sum_{n=-\infty}^{\infty} |h[n]e^{-j\theta n}| \\
&= \sum_{n=-\infty}^{\infty} \left| \sqrt{(h[n]e^{-j\theta n})(h^*[n]e^{j\theta n})} \right| \\
&= \sum_{n=-\infty}^{\infty} |h[n]|.
\end{aligned}$$

ROC contains the unit circle $|z|=1$ $\longleftrightarrow$ $h[n]$ is absolutely summable

$\Updownarrow$

BIBO Stable

An LTI system is BIBO stable if and only if the ROC of its transfer function contains $|z|=1$, i.e. the unit circle

$|z| > \frac{1}{2}$ ✓

$|z| < \frac{1}{2}$ ✗

$|z| < 3$ ✓

$|z| > 1$ ✗

# Relating BIBO Stability, Causality, and $z$-domain

Recall that infinite-length signals can fall into one of three categories:

1. Right-sided: $x[n] = 0$, $n < n_0$, $n_0 \in \mathbb{Z}$
   - The ROC of right-sided signals is $|z| > a$, $a > 0$.
2. Left-sided: $x[n] = 0$, $n > n_0$, $n_0 \in \mathbb{Z}$
   - The ROC of left-sided signals is $|z| < b$, $b > 0$.
3. Two-sided: $x[n]$ is nonzero for $-\infty < n < \infty$.
   - The ROC of two-sided signals is $a < |z| < b$, $0 < a < b$.



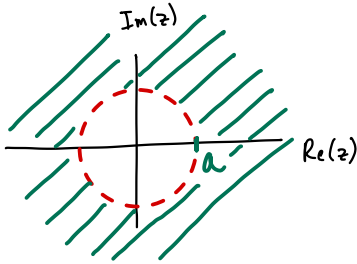
# Relating BIBO Stability, Causality, and $z$-domain: Right-sided Signals

- Right-sided: $x[n] = 0,\ n < n_0,\ n_0 \in \mathbb{Z}$
  - The ROC of right-sided signals is $|z| > a,\ a > 0$.

$a^n u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{1-az^{-1}},\ |z| > |a|$

$a = |p_{max}|$ = largest magnitude → finite pole

Right-sided systems are BIBO stable iff $|a| < 1$




- Finite poles are poles not at $z = 0$ or $z = \pm\infty$

→ Right-sided systems have past feedback

Ex: $y[n] = y[n-1] + x[n]$.

Two Cases:

① $n_0 \geq 0$: Causal signals/systems

Ex: $h[n] = \left(\frac{1}{2}\right)^{n-3} u[n-3]$

$n_0 = 3$ $\quad H(z) = \frac{z^{-3}}{1-\frac{1}{2}z^{-1}},\ |z| > \frac{1}{2}$

② $n_0 < 0$: Non-causal

Ex: $h[n] = u[n+1] = \delta[n+1] + u[n]$

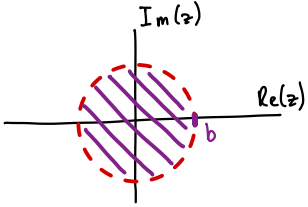
$n_0 = -1$ $\quad H(z) = z + \frac{1}{1-z^{-1}},\ 1 < |z| < \infty$

↑ new pole @ $|z| = \infty$

# Relating BIBO Stability, Causality, and $z$-domain: Left-sided Signals

- Left-sided: $x[n] = 0,\ n > n_0,\ n_0 \in \mathbb{Z}$
  - The ROC of left-sided signals is $|z| < b,\ b > 0$.






$-b^n u[-n-1] = -b^n u[-(n+1)] \longleftrightarrow \frac{1}{1-bz^{-1}},\ |z| < |b|$

$b = |p_{min}|$ = smallest magnitude finite pole → Left-sided system are BIBO stable iff $|b| > 1$

→ Left-sided systems have future feedback terms

→ Ex: $y[n] = 3y[n+1] + x[n+1]$ (go backwards through $x[n]$)

## Two Cases

① $n_0 \geq 0$: Non-causal

Ex: $h[n] = -3^{n-3} u[-(n-2)]$

$n_0 = 2$

$H(z) = \frac{z^{-3}}{1-3z^{-1}},\ 0 < |z| < 3$ (new pole @ $z = 0$)

② $n_0 < 0$: Anti-causal → depend only on future input or feedback terms

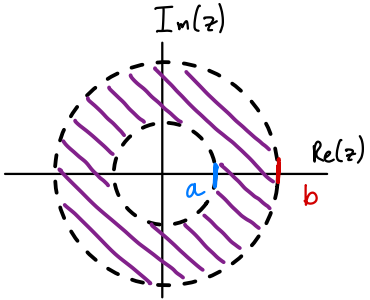
$h[n] = -3^{n+2} u[-(n+3)]$

$n_0 = -3$

$H(z) = \frac{z^2}{1-3z^{-1}},\ |z| < 3$

# Relating BIBO Stability, Causality, and $z$-domain: Two-sided Signals

- Two-sided: $x[n]$ is nonzero for $-\infty < n < \infty$.
  - The ROC of two-sided signals is $a < |z| < b$, $0 < a < b$.




$a = |p_{max,\, x_R}|$

$b = |p_{min,\, x_L}|$

- Let $x[n] = x_L[n] + x_R[n]$ where

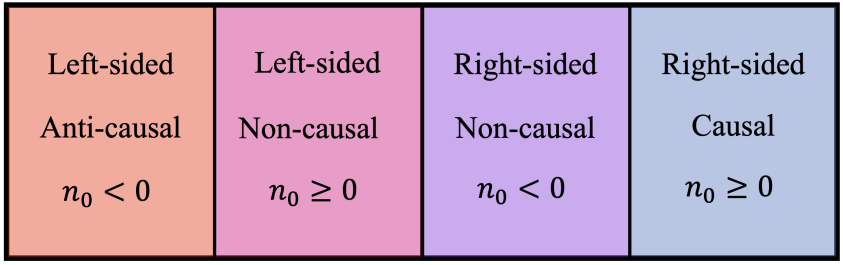
$$x_L[n] = \begin{cases} x[n], & n < 0 \\ 0, & n \geq 0 \end{cases}$$

$$x_R[n] = \begin{cases} 0, & n < 0 \\ x[n], & n \geq 0 \end{cases}$$

Stable iff $a < 1$ and $b > 1$.

- ROC of $x[n] = ROC_{x_L} \cap ROC_{x_R}$
- $ROC_{x_L} = |z| < |p_{min,\, x_L}|$, $ROC_{x_R} = |z| > |p_{max,\, x_R}|$

# Relating BIBO Stability, Causality, and $z$-domain: Summary




| ROC shape of $H(z)$ | Causality Type | Right-sided | Left-sided | Condition for stability |
|---|---|---|---|---|
| $\lvert z\rvert > \lvert p_{\max}\rvert$ | Causal | ✓ | ✗ | $\lvert p_{\max}\rvert < 1$ |
| $\lvert z\rvert < \lvert p_{\min}\rvert$ | Anti-causal | ✗ | ✓ | $\lvert p_{\min}\rvert > 1$ |
| $\lvert p_{\max}\rvert < \lvert z\rvert < \infty$ | Non-causal | ✓ | ✗ | $\lvert p_{\max}\rvert < 1$ |
| $0 < \lvert z\rvert < \lvert p_{\min}\rvert$ | Non-causal | ✗ | ✓ | $\lvert p_{\min}\rvert > 1$ |
| $a < \lvert z\rvert < b$ | Non-causal | ✓ | ✓ | $\lvert p_{\max,h_r}\rvert = a < 1$ and $\lvert p_{\min,h_l}\rvert = b > 1$ |

# Example: BIBO Stability of IIR Systems

For each LTI system, determine the corresponding $h[n]$ for which the system is BIBO stable. Also state if this stable system is causal, non-causal, or anti-causal.

$$H(z) = \frac{1}{1-\frac{1}{2}z^{-1}} + \frac{1}{1+\frac{4}{3}z^{-1}}$$

Poles @: $p_1 = \frac{1}{2}$, $p_2 = -\frac{4}{3}$

$(p_1, p_2) = (RS, RS)$: ROC $= |z| > \frac{1}{2} \cap |z| > \frac{4}{3}$

$= |z| > \frac{4}{3}$ ✗

$(p_1, p_2) = (RS, LS)$: ROC $= |z| > \frac{1}{2} \cap |z| < \frac{4}{3}$

$= \frac{1}{2} < |z| < \frac{4}{3}$ ✓

$h[n] = \left(\frac{1}{2}\right)^n u[n] - \left(-\frac{4}{3}\right)^n u[-n-1]$ → non-causal ✓

$$H(z) = \frac{1}{1-\frac{1}{5}z^{-1}} + \frac{1}{1-\frac{4}{5}z^{-1}}$$

Poles @: $p_1 = \frac{1}{5}$, $p_2 = \frac{4}{5}$

$(p_1, p_2) = (RS, RS)$: ROC $= |z| > \frac{1}{5} \cap |z| > \frac{4}{5}$

$= |z| > \frac{4}{5}$ ✓

→ $h[n] = \left(\frac{1}{5}\right)^n u[n] + \left(\frac{4}{5}\right)^n u[n]$ → Causal

Causal: $a^n u[n] \xleftrightarrow{\mathcal{Z}} \frac{1}{1-az^{-1}}$, $|z| > |a|$

Anti-causal: $-b^n u[-n-1] \xleftrightarrow{\mathcal{Z}} \frac{1}{1-bz^{-1}}$, $|z| < |b|$

# Example: FIR System Stability

Can an LTI FIR system be unstable? Justify your reasoning in the time-domain and $z$-domain.

Consider this question while getting ready for class!

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# Example: FIR System Stability

Can an LTI FIR system be unstable? Justify your reasoning in the time-domain and $z$-domain.

- Let $h[n] = \{b_0, b_1, b_2, \cdots, b_{N-1}\}$ be length-$N$. Thus,

$y[n] = \sum_{k=0}^{N-1} b_k x[n-k]$ represents this length-$N$ FIR system.

## Time-domain:

Let $\max_n \{h[n]\} = b_{max}$.

$$\sum_{n=-\infty}^{\infty} |h[n]| = \sum_{n=0}^{N-1} |h[n]|$$
$$\leq \sum_{n=0}^{N-1} b_{max}$$
$$= N b_{max} < \infty \quad \checkmark$$

## z-domain:

$$y[n] = \sum_{k=0}^{N-1} b_k x[n-k] \xleftrightarrow{\mathcal{Z}} Y(z) = X(z) \sum_{k=0}^{N-1} b_k z^{-k}$$

$$\frac{Y(z)}{X(z)} = H(z) = \frac{\sum_{k=0}^{N-1} b_k z^{-1}}{1}$$

→ No finite poles

→ ROC = all $z$ (except possibly $z=0$ or $z=\infty$)

↳ Contains $|z|=1$ ✓

# Unstable Systems and Unstable Inputs

When a system is unstable, it is important to identify what (bounded) input signals may lead to an unbounded output. Consider, for example, an unstable LTI system with impulse response $h[n] = 3^n u[n]$. What bounded inputs yield unbounded outputs for this system?

$$H(z) = \frac{1}{1-3z^{-1}}, \quad |z| > 3 \rightarrow \text{Unstable!}$$

Ex: $x[n] = u[n]$

$$X(z) = \left(\frac{1}{3}\right)^n u[n]$$

Almost any $x[n]$ produces an unbounded output:

$x[n] = \delta[n] \rightarrow y[n] = x[n] * h[n] = \delta[n] * h[n] = 3^n u[n]$

# Unstable Systems and Unstable Inputs

Consider, for example, an unstable LTI system with impulse response $h[n] = 3^n u[n]$. What bounded inputs also yield bounded outputs for this system?

$$H(z) = \frac{1}{1-3z^{-1}}, \quad |z| > 3$$

Must cancel pole @ $z=3$:

Ex: $X(z) = 1-3z^{-1} \xleftrightarrow{\mathcal{Z}} x[n] = \delta[n] - 3\delta[n-1]$

$$Y(z) = X(z)H(z) = (1-3z^{-1})\left(\frac{1}{1-3z^{-1}}\right) = 1 \xleftrightarrow{\mathcal{Z}} \delta[n] = y[n]$$

# Unstable Systems and Unstable Inputs

Suppose instead we have an unstable LTI system with the following impulse response.

$$h[n] = (-1)^n u[n]$$

$$H(z) = \frac{1}{1+z^{-1}}, \quad |z| > 1 \rightarrow \text{Unstable!}$$

$$x[n] = \delta[n]$$

$$y[n] = x[n] * h[n]$$

$$= \delta[n] * h[n] = (-1)^n u[n] \rightarrow \text{Bounded output!}$$



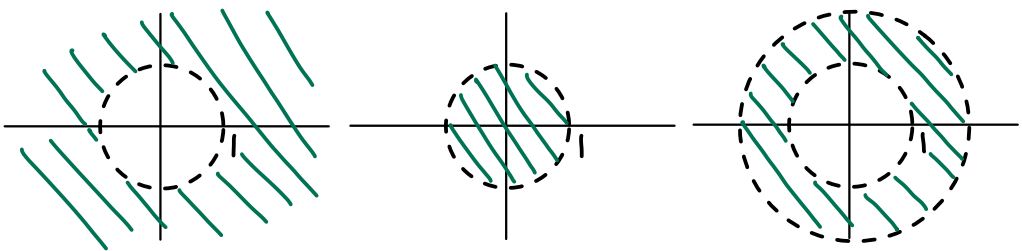
# Marginal Stability

## Marginal Stability

An LTI system with impulse response $h[n]$ and transfer function $H(z)$ is *marginally stable* if its ROC is open on the unit circle, i.e. $|z| < 1$ or $|z| > 1$.

What bounded inputs produce unbounded outputs for marginally stable systems?

Example marginally stable regions of convergence:



# Pole Matching: Time-Domain Interpretation

A causal marginally stable system can be described by an impulse response $h[n] = a^n u[n]$, for $|a| = 1, a \in \mathbb{C}$. Let $x[n] = b^n u[n]$. When is $y[n] = x[n] * h[n]$ an unbounded output?

Three cases to consider:

(a) $|b| > 1$

(b) $|b| < 1$

(c) $|b| = 1$



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(a) $|b| > 1$:

$x[n] = b^n u[n]$

→ $x[n]$ is an unbounded signal, i.e. $x[n] \to \infty$ as $n \to \infty$

→ Not valid for checking BIBO stability.

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Three cases to consider:

(a) $|b| > 1$

(b) $|b| < 1$

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(a) $|b| > 1$:

$X[n] = b^n u[n]$

→ $X[n]$ is an unbounded signal, i.e. $x[n] \to \infty$ as $n \to \infty$

→ Not valid for checking BIBO stability.

(b) $|b| < 1$:

$$Y(z) = X(z)H(z) = \underbrace{\left(\frac{1}{1-bz^{-1}}\right)}_{|z|>|b|}\underbrace{\left(\frac{1}{1-az^{-1}}\right)}_{|z|>1}, \quad |z| > 1$$

Goes to zero as $n \to \infty$

Stays at magnitude 1 for all $n$

$$Y(z) = \frac{1}{(1-bz^{-1})(1-az^{-1})} = \frac{A_1}{1-bz^{-1}} + \frac{A_2}{1-az^{-1}} \xrightarrow{\mathcal{Z}^{-1}} y[n] = A_1(b)^n u[n] + A_2(a)^n u[n]$$

→ Bounded!

# Pole Matching: Time-Domain Interpretation

A causal marginally stable system can be described by an impulse response $h[n] = a^n u[n]$, for $|a| = 1, a \in \mathbb{C}$. Let $x[n] = b^n u[n]$. When is $y[n] = x[n] * h[n]$ an unbounded output?

(c) If $|a| = 1$ and $|b| = 1$, then let $a = e^{j\theta}$ and $b = e^{j\phi}$:



# Pole Matching: Time-Domain Interpretation

A causal marginally stable system can be described by an impulse response $h[n] = a^n u[n]$, for $|a| = 1, a \in \mathbb{C}$. Let $x[n] = b^n u[n]$. When is $y[n] = x[n] * h[n]$ an unbounded output?

(c) If $|a| = 1$ and $|b| = 1$, then let $a = e^{j\theta}$ and $b = e^{j\phi}$:

Claim: $y[n]$ is unbounded iff $\theta = \phi$.

# Pole Matching: Time-Domain Interpretation

A causal marginally stable system can be described by an impulse response $h[n] = a^n u[n]$, for $|a| = 1, a \in \mathbb{C}$. Let $x[n] = b^n u[n]$. When is $y[n] = x[n] * h[n]$ an unbounded output?

Claim: $y[n]$ is unbounded iff $\theta = \varphi$.

(c) If $|a| = 1$ and $|b| = 1$, then let $a = e^{j\theta}$ and $b = e^{j\phi}$:

$$y[n] = x[n] * h[n]$$

$$= \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

$$= \sum_{k=-\infty}^{\infty} e^{j\theta k} u[k]\, e^{j\varphi(n-k)} u[n-k]$$

$$= \sum_{k=0}^{n} e^{j\theta k} e^{j\varphi n} e^{-j\varphi k}$$

$$= e^{j\varphi n} \sum_{k=0}^{n} e^{j(\theta-\varphi)k}$$

# Pole Matching: Time-Domain Interpretation

A causal marginally stable system can be described by an impulse response $h[n] = a^n u[n]$, for $|a| = 1, a \in \mathbb{C}$. Let $x[n] = b^n u[n]$. When is $y[n] = x[n] * h[n]$ an unbounded output?

Claim: $y[n]$ is unbounded iff $\theta = \varphi$.

(c) If $|a| = 1$ and $|b| = 1$, then let $a = e^{j\theta}$ and $b = e^{j\phi}$:

$$y[n] = x[n] * h[n]$$

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$$= \sum_{k=-\infty}^{\infty} e^{j\theta k} u[k]\, e^{j\varphi(n-k)} u[n-k]$$

$$= \sum_{k=0}^{n} e^{j\theta k} e^{j\varphi n} e^{-j\varphi k}$$

$$= e^{j\varphi n} \sum_{k=0}^{n} e^{j(\theta-\varphi)k}$$

If $\theta = \varphi$: $= e^{j\varphi n} \sum_{k=0}^{n} 1 = e^{j\varphi n}(n+1)\, u[n]$ → Unbounded!

If $\theta \neq \varphi$: $= e^{j\varphi n} \cdot \frac{1 - e^{j(\theta-\varphi)n+1}}{1 - e^{j(\theta-\varphi)}} u[n]$ → Bounded! (oscillates)

# Pole Matching: Time-Domain Interpretation

A causal marginally stable system can be described by an impulse response $h[n] = a^n u[n]$, for $|a| = 1, a \in \mathbb{C}$. Let $x[n] = b^n u[n]$. When is $y[n] = x[n] * h[n]$ an unbounded output?

(c) If $|a| = 1$ and $|b| = 1$, then let $a = e^{j\theta}$ and $b = e^{j\phi}$:

Claim: $y[n]$ is unbounded iff $\theta = \varphi$.

$$y[n] = x[n] * h[n]$$

$$= \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

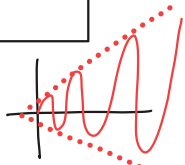
$$= \sum_{k=-\infty}^{\infty} e^{j\theta k} u[k]\, e^{j\varphi(n-k)} u[n-k]$$

$$= \sum_{k=0}^{n} e^{j\theta k} e^{j\varphi n} e^{-j\varphi k}$$

$$= e^{j\varphi n} \sum_{k=0}^{n} e^{j(\theta-\varphi)k}$$

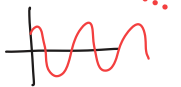
$\theta = \varphi$:

$$= e^{j\varphi n} \sum_{k=0}^{n} 1 = e^{j\varphi n}(n+1)\, u[n] \rightarrow \text{Unbounded!}$$



$\theta \neq \varphi$:

$$= e^{j\varphi n} \cdot \frac{1 - e^{j(\theta-\varphi)n+1}}{1 - e^{j(\theta-\varphi)}}\, u[n] \rightarrow \text{Bounded! (oscillates)}$$



Conclusion: Marginally stable systems only produce unbounded outputs if the input signal matches at least one pole of the system on the unit circle.

# Example: Pole Matching

Consider a causal LTI system with the following transfer function:

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Poles e: $z = j$, $z = -1$, $z = \frac{2}{3}$

$$H(z) = \frac{1}{(1 - jz^{-1})(1 + z^{-1})(1 - \frac{2}{3}z^{-1})}. \quad , |z| > |$$

Which of the following input signals will lead to an unbounded output?

1. $(-1)^n u[n]$ → Poles e $z = -1$
2. $u[n]$ → Poles e $z = 1$
3. $e^{j\frac{\pi}{2}n} u[n]$ → Poles e $z = e^{j\frac{\pi}{2}} = j$
4. $e^{-j\frac{\pi}{2}n} u[n]$ → Poles e $z = e^{-j\frac{\pi}{2}} = -j$
5. $\left(\frac{2}{3}\right)^n u[n]$ → Poles e $z = \frac{2}{3}$
6. $\sin\left(-\frac{\pi}{2}n\right) u[n]$ → Poles e $z = \pm j = e^{\pm j\frac{\pi}{2}}$
7. $\cos(\frac{\pi}{2}n) u[n]$ → Poles e $z = \pm j = e^{\pm j\frac{\pi}{2}}$
8. $\sin\left(\frac{2\pi}{3}n\right) u[n]$ → Poles e $z = e^{\pm j\frac{2\pi}{3}}$

$$a^n u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{1 - az^{-1}}, |z| > |a|$$

$$\cos(\theta_0 n) u[n] = \frac{1}{2} e^{j\theta_0 n} u[n] + \frac{1}{2} e^{-j\theta_0 n} u[n]$$

$$\sin(\theta_0 n) u[n] = \frac{1}{2j} e^{j\theta_0 n} u[n] - \frac{1}{2j} e^{-j\theta_0 n} u[n]$$

# Example: Pole Matching

Consider a causal LTI system with the following transfer function:

$$H(z) = \frac{1}{(1 - jz^{-1})(1 + z^{-1})(1 - \frac{2}{3}z^{-1})}.$$

Which of the following input signals will lead to an unbounded output?

1. $(-1)^n u[n]$
2. $u[n]$
3. $e^{j\frac{\pi}{2}n}u[n]$
4. $e^{-j\frac{\pi}{2}n}u[n]$
5. $\left(\frac{2}{3}\right)^n u[n]$ → Produce a double pole in $Y(z)$ at $z = \frac{2}{3}$. This leads to an $n\left(\frac{2}{3}\right)^n u[n]$ term in $y[n]$. → Bounded.
6. $\sin\left(-\frac{\pi}{2}n\right) u[n]$
7. $\cos(\frac{\pi}{2}n)u[n]$
8. $\sin\left(\frac{2\pi}{3}n\right) u[n]$

$\frac{1}{2}e^{j\frac{\pi}{2}n}u[n] \overset{\mathcal{Z}}{\leftrightarrow} \frac{\frac{1}{2}}{1 - e^{j\frac{\pi}{2}}z^{-1}}, \quad |z| > 1$